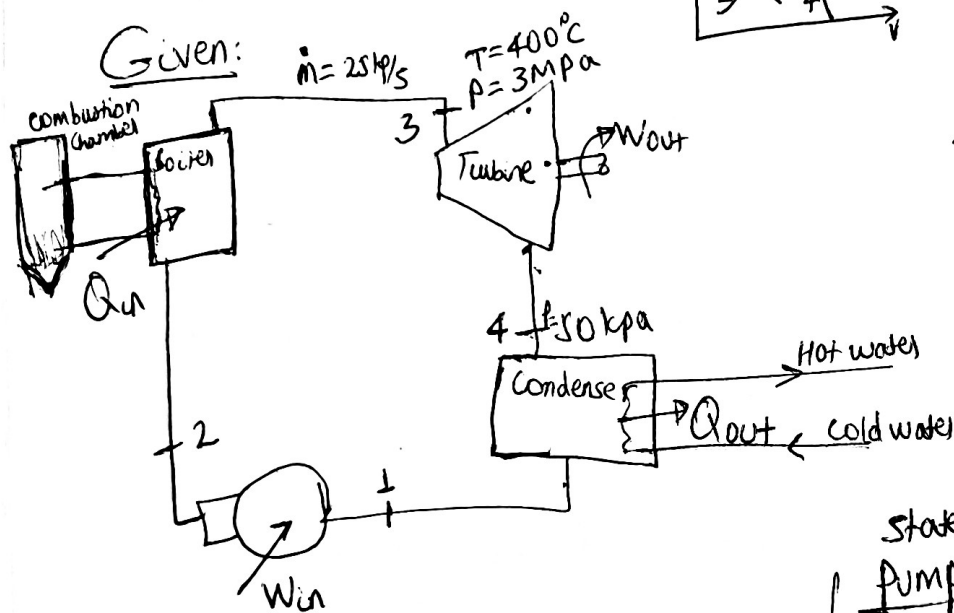


4. Steam power plant : Rankine cycle ①

A steam power plant operates on a simple ideal Rankine cycle between the pressure limits of 3 MPa & 50 kPa. The temperature of the steam at the turbine inlet is 400°C, and the mass flow rate of steam through the cycle is 25 kg/s. Show the cycle on a T-s diagram with respect to saturation lines, and determine

- the thermal efficiency of the cycle and
- the net power output of the powerplant.

Solution



Assumptions

- Steady operating conditions exist.
- Kinetic energy & potential energy changes are negligible.

Analysis

- Start with conditions you know.

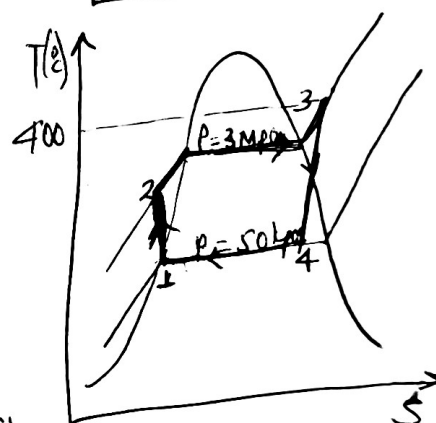
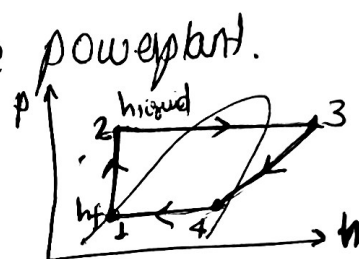
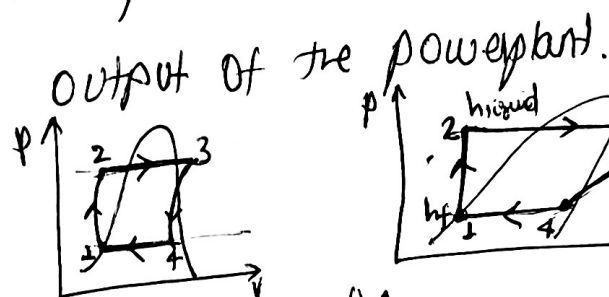
State 1 : Saturated liquid

$P = 50 \text{ kPa}$

Steam pressure table : Table A-4

$h_1 = 340.49 \text{ kJ/kg}$

$v_1 = 0.001030 \text{ m}^3/\text{kg}$



State 1 $\rightarrow$ 2

Pump energy balance

$$W_{\text{pump}} = v_1 (P_2 - P_1) = h_2 - h_1$$

$$\begin{aligned} W_{\text{out}} - W_{\text{in}} + Q_{\text{in}} &= \Delta H \\ -W_{\text{in}} &= \Delta h \\ -(P(v_2 - v_1)) &= \Delta u \end{aligned}$$

$$W_p = \int v dp = v \Delta P$$

Note:

U (internal energy)

- Energy required to create the system.

H (Enthalpy)

- Energy required to make the created system flow.

$\hookrightarrow$ internal energy + flow work.

① For flowing fluid use $\Delta H/\dot{m}$ instead of ΔU .

②

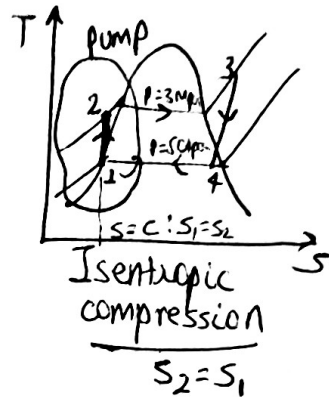
$$\Rightarrow W_{\text{pump}} = h_2 - h_1 = V_1 (P_2 - P_1)$$

$$= 0.001030 \text{ m}^3/\text{kg} * (3000 \text{ kPa} - 50 \text{ kPa})$$

$$= 0.001030 \text{ m}^3/\text{kg} * 250 \text{ kPa}$$

$$= 3.0385 \text{ m}^3/\text{kg} * \cancel{\text{kPa}} * \cancel{\text{kJ/m}^3} / \cancel{\text{kPa}}$$

$$= 3.0385 \text{ kJ/kg}$$



$$W_{\text{pump}} = \underline{3.0385 \text{ kJ/kg}}$$

From above $h_2 = W_{\text{pump}} + h_1$

$$= 3.0385 \text{ kJ/kg} + 340.49 \text{ kJ/kg}$$

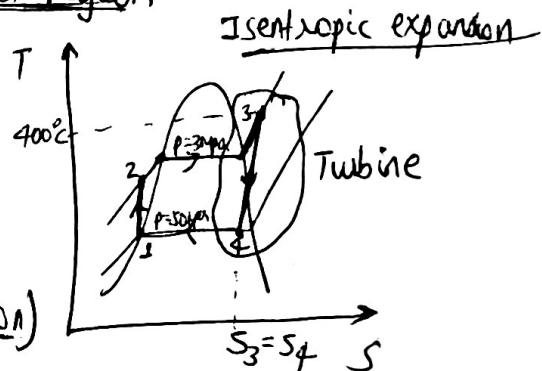
$$= \underline{343.529 \text{ kJ/kg}}$$

State 3: $P = 3 \text{ MPa}$, $T = 400^\circ\text{C} \Rightarrow$ superheated region

Table: A-4

State 3: $h_3 = 3230.9 \text{ kJ/kg}$

$s_3 = 6.9212 \text{ kJ/kg} \cdot \text{K}$



State 4 Mixture region (sat. liquid + sat. vapor region)

From $s_3 = s_4 = 6.9212 \text{ kJ/kg} \cdot \text{K}$

$$s_4 = s_f + X s_{fg}$$

s_f from table A-4 with $P = 50 \text{ kPa}$ $s_f = 1.0912 \text{ kJ/kg} \cdot \text{K}$

$s_{fg} = 6.5019 \text{ kJ/kg} \cdot \text{K}$

$$\Rightarrow s_4 = s_f + X s_{fg}$$

$$6.9212 \text{ kJ/kg} \cdot \text{K} = 1.0912 \text{ kJ/kg} \cdot \text{K} + X (6.5019 \text{ kJ/kg} \cdot \text{K})$$

$$X = X_4 = \underline{0.8966}$$

$$\Rightarrow h_4 = h_f + X h_{fg}$$

$$= 340.54 \text{ kJ/kg} + 0.8966 * 2304.7 \text{ kJ/kg}$$

$$= \underline{2406.9 \text{ kJ/kg}}$$

Table A-4: $P = 50 \text{ kPa}$

$$h_f = 340.54 \text{ kJ/kg}$$

$$h_{fg} = 2304.7 \text{ kJ/kg}$$

a. η_{th}

$$\eta_{th} = \frac{W_{net}}{Q_{in}}$$

And $W_{net} = W_{turb} - W_{pump}$
 $= (h_3 - h_4) - (h_2 - h_1)$

$$= 823.49 - 3.039 \text{ kJ/kg}$$

$$W_{net} = \underline{\underline{820.45 \text{ kJ/kg}}}$$

Q_{in} : Boiler : state 2 - 3

$$Q_{in} = h_3 - h_2 = \underline{\underline{2887.37 \text{ kJ/kg}}}$$

$$\therefore \eta_{th} = \frac{820.45 \text{ kJ/kg}}{2887.37 \text{ kJ/kg}} = \underline{\underline{28.4 \%}}$$

Since the turbine output is in the mixture region, η_{th} will be less.
 (it loses efficiency)

b. Power Output: From $\dot{W}_{net} = \dot{m} * W_{net}$

$$\boxed{W_{net} = \frac{\dot{W}_{net}}{\dot{m}}} \Rightarrow \dot{W}_{net} = 25 \text{ kg/s} * 820.45 \text{ kJ/kg}$$

$$= \underline{\underline{20.511 \text{ MW}}}$$

